

Lecture 24

Saturday, May 9, 2026 9:04 AM

Today: Principal Component Analysis (PCA)

- Recap of key linear algebra concepts
- What is the goal of PCA?
- PCA as an optimization problem & its solution
- Examples

$$\rightarrow \begin{bmatrix} a & b & c & d \\ e & f & g & h \\ i & j & k & l \\ m & n & o & p \end{bmatrix}$$

$$\begin{aligned} & a \det \begin{pmatrix} \begin{bmatrix} f & g & h \\ j & k & l \\ n & o & p \end{bmatrix} \end{pmatrix} - b \det \begin{pmatrix} \begin{bmatrix} e & g & h \\ i & k & l \\ m & o & p \end{bmatrix} \end{pmatrix} \\ & + c \det \begin{pmatrix} \begin{bmatrix} e & f & h \\ i & j & l \\ m & n & p \end{bmatrix} \end{pmatrix} - d \det \begin{pmatrix} \begin{bmatrix} e & f & g \\ i & j & k \\ m & n & o \end{bmatrix} \end{pmatrix} \end{aligned}$$

Computing eigenvalues or eigenvector

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \quad v = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} \quad \lambda$$

$$\begin{aligned} \text{RHS} = A v &= \lambda v \quad \text{--- (1)} \\ \lambda \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} &= \begin{bmatrix} \lambda v_1 \\ \lambda v_2 \end{bmatrix} = \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} \\ &= \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} \end{aligned}$$

* tricky

$$\begin{aligned} \text{(1)} &\Rightarrow (A - \lambda I) v = 0 \\ &\Rightarrow \begin{pmatrix} a-\lambda & b \\ c & d-\lambda \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = 0 \quad \text{--- (2)} \end{aligned}$$

$$\text{(2) is possible only if } \det \begin{pmatrix} a-\lambda & b \\ c & d-\lambda \end{pmatrix} = 0$$

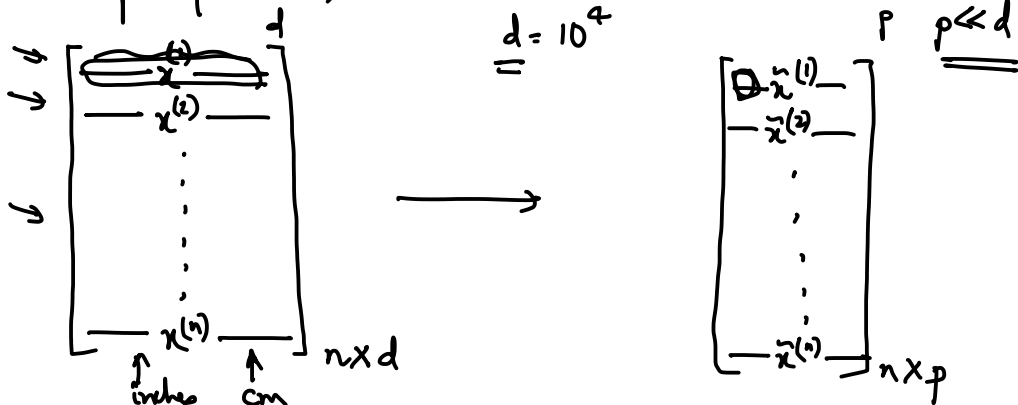
$$\Rightarrow (a-\lambda)(d-\lambda) - bc = 0$$

solve this to obtain λ , use λ to obtain v

$A : m \times m \Rightarrow$ polynomial of degree m
 $\Rightarrow$ at most m roots

PCA

n sample points, each is d dimension

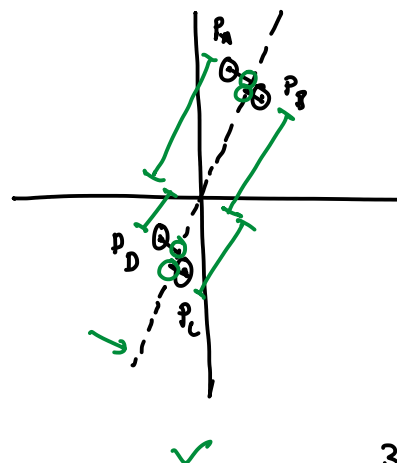
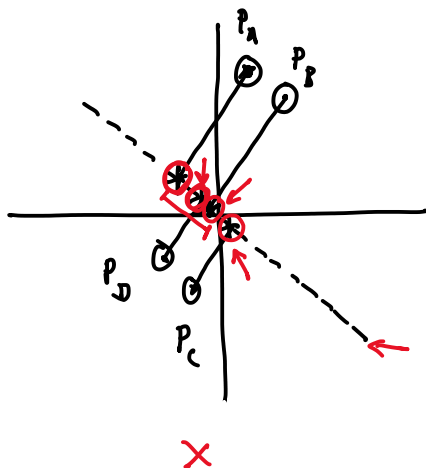


- computation / storage
- visualization
- redundancy

$d=2$ $p=1$

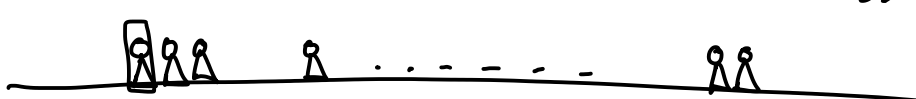
$\rightarrow$ PCA : Transform data from $\mathbb{R}^d$ to $\mathbb{R}^p$ such that the variance in the data is maximized.

$n=4$



3D

$C_1 \rightarrow$



2D

$\uparrow C_2$

PCA (1, 1, 1, 1)

PCA (Trusnea)

- n data points $x_1, \dots, x_n \in \mathbb{R}^d$

- maximize the variance of the projection

$$\max_{u_1} u_1^T S u_1 \quad \text{s.t.} \quad \|u_1\| = 1$$

 u_1 is unit vector S = covariance matrix of the data

↑

- When you solve the problem (spectral theorem)
 u_1 - eigenvector of S corresponding to the largest eigenvalue

Proof omitted

